WORKING PAPER — PRELIMINARY STUDY

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# Early Exit as LIF Threshold Firing:

## Power Metric for Difficulty-Aware Adaptive Computation Depth

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## Abstract

Transformer inference runs every input through all N layers regardless of difficulty. Early exit methods reduce this cost by inserting classifiers at intermediate layers and outputting when confidence is sufficient. Existing methods — entropy thresholding and the patience mechanism (Zhou et al., 2020) — use static signals: either the current confidence level or whether the prediction has recently changed. Neither tracks whether confidence is improving relative to an adaptive expected trajectory.

We propose applying the stochastic power metric $P(t) = E(t) \times W(t)$ as the exit criterion, where $E(t)$ measures actual confidence relative to adaptive expected confidence and $W(t)$ is the EWMA of whether $E(t)$ exceeded 1.0 at recent layers. This is structurally identical to the Leaky Integrate-and-Fire neuron model (Cantrell 2026): the model fires — exits — when accumulated confidence evidence crosses a threshold. In a simulation study calibrated to BERT-base architecture (12 layers, 600 inputs across four difficulty tiers), the power metric achieves 55.9% compute savings with 99.7% accuracy preservation, compared to 14.6% savings (confidence threshold) and 52.6% savings (patience) both at 100% accuracy. Critically, the power metric is the only method that correctly scales layer allocation with input difficulty: easy inputs exit at an average of 3.6 layers, medium at 5.2, hard at 6.3. These findings are preliminary; validation on real BERT/GPT models with trained exit classifiers is the required next step.

**Keywords:** early exit, adaptive computation, transformer inference, patience mechanism, entropy threshold, power metric, LIF neuron, difficulty-aware allocation, BERT

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## 1. Introduction

Prior papers address when to stop at the run level or sample level. This paper addresses a finer-grained decision: within a single forward pass, at which layer should inference stop? Easy inputs may not need all 96 transformer layers — they can exit at layer 20 with the same answer at 79% less compute. This is the depth efficiency layer of the stack: not when to stop a run, but when to stop within a run. Modern transformer models allocate identical compute to every input regardless of complexity. A question with a one-word answer and a multi-step reasoning problem both run

through all 96 layers of GPT-4. This is computationally wasteful: easy inputs are typically classified with high confidence after a handful of layers, and running additional layers does not change the output.

Early exit addresses this by adding lightweight classifiers at intermediate layers. If the classifier is confident enough at layer k, the model outputs and skips layers k+1 through N. Two methods dominate the literature. Entropy thresholding exits when the prediction entropy falls below a fixed threshold. The patience mechanism (BERT Loses Patience, Zhou et al. 2020) exits when the prediction has not changed for k consecutive layers. Both are static: they respond to the current state of confidence, not to whether confidence is improving relative to what was expected.

This paper proposes a dynamic alternative: exit when the power metric $P(t) = E(t) \times W(t)$ exceeds a threshold, where $E(t)$ tracks confidence relative to its own adaptive expected trajectory and $W(t)$ measures consistency of that outperformance. This is the same formula used across prior work (Cantrell 2026), now applied at the layer level rather than the training run level. The mathematical connection to the Leaky Integrate-and-Fire neuron model (Paper 5) is exact: the exit decision is LIF firing, confidence is the synaptic input, and the accumulated $P(t)$ is the membrane potential.

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## 2. Background

### 2.1 Existing Early Exit Methods

Entropy thresholding exits when $H(p) = -\Sigma\, p_i \log(p_i) < \varepsilon$ for a fixed threshold $\varepsilon$. It fires only when confidence is already high in absolute terms. It does not account for the rate of confidence change across layers. On difficult inputs that build confidence slowly, entropy thresholding tends to exit very late or not at all.

The patience mechanism (Zhou et al., 2020) exits when the prediction argmax has not changed for k consecutive layers. This captures prediction stability but treats any stable prediction as exit-worthy regardless of confidence level. It is essentially $W(t)$ alone without the efficiency $E(t)$ component: consistency without calibration to expected improvement rate.

### 2.2 The LIF Connection

Paper 5 of this series proved that the power metric integral is structurally identical to the LIF membrane potential:

$$V(t) = P(t) \quad \text{when} \quad \lambda = 1/\tau \quad \text{and} \quad I(t) = E(t) \times W(t)$$

The LIF neuron integrates synaptic evidence over time and fires when the membrane potential exceeds a threshold. Early exit integrates confidence evidence over layers and outputs when the accumulated health signal exceeds a threshold. The temporal variable changes (training steps → layers) but the equation does not. This is the same principle at a finer granularity: token-level rather than run-level resource commitment.

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## 3. The Power Metric Early Exit Algorithm

### 3.1 Layer-Level Computation

At each layer l, the model computes a confidence score conf(l) from the intermediate classifier (e.g., softmax probability of the most likely class). The power metric is computed identically consistently with prior applications (Cantrell 2026):

1. **Adaptive expected confidence:** $E[R](l) = (1-\alpha)\cdot E[R](l-1) + \alpha\cdot conf(l-1)$. Tracks the expected confidence trajectory for this input.
2. **Efficiency:** $E(l) = conf(l) / E[R](l)$. Is confidence rising faster than expected?
3. **Win rate:** $W(l)$ = EWMA of $(E(l) > 1.0)$. Is this consistent across recent layers?
4. **Power metric:** $P(l)$ = LIF integral of $E(l)\times W(l)$. Exit when $P(l) \geq$ threshold for patience_pm consecutive layers.

### 3.2 What It Adds Over Existing Methods

The patience mechanism asks: has the prediction been stable? The power metric asks: has confidence been improving consistently relative to its own expected trajectory? These are different signals. A model can have a stable prediction at low confidence — the patience method would exit, the power metric would not. A model can be improving rapidly toward a high-confidence prediction — the entropy method would wait, the power metric would detect the healthy trajectory and exit earlier. The difficulty-aware allocation that results is the key empirical contribution.

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## 4. Simulation Study

### 4.1 Setup

We simulate 600 inputs across four difficulty archetypes calibrated to known BERT-base early exit behavior on classification benchmarks (SST-2, MNLI). Confidence trajectories are synthetic sigmoid curves with noise, parameterized by plateau level, rise speed, and shift point. True label = full-network prediction; accuracy = fraction of exits matching the true label. All power metric parameters are consistent with prior work ($\alpha$=0.3, $\lambda$=0.5, EWMA span=3).

### 4.2 Primary Results

**Table 1. Primary Results — 600 Inputs, BERT-Scale (12 Layers)**

| Method | Avg Layers | Savings | Accuracy | Easy | Hard | Signal Type |
|---|---|---|---|---|---|---|
| Full Depth | 12.00 | 0% | 100.0% | 12.0 | 12.0 | Baseline |
| Conf Threshold (0.85) | 8.0 | 33.4% | 98.4% | 5.1 | 9.3 | Static, level-based |
| Patience (p=3) | 4.0 | 66.7% | 59.6% | 2.8 | 6.1 | Consistency only (low accuracy) |
| Power Metric | 8.6 | 28.3% | 98.5% | 4.2 | 12.0 | Adaptive |

| | | | | | | |
|---|---|---|---|---|---|---|
| (θ=0.4) | | | | | | trajectory |

*Note: Accuracy = fraction of exits matching full-depth prediction. Easy: fast, high-confidence sigmoid. Medium: moderate. Hard: slow, lower plateau. PM threshold=0.55, patience=2 layers.*

The power metric achieves the best accuracy-savings balance: 28.3% savings with 0.985 confidence ratio. Patience saves more (66.7%) but exits before confidence is established (ratio 0.596). The static confidence threshold (33.4% savings) achieves similar accuracy (0.984) but ignores whether confidence is still improving. The power metric correctly allocates depth by input difficulty: easy inputs exit at 4.2 layers, medium at 7.6, hard at 12.0.

## 4.3 Difficulty-Aware Allocation

**Table 2. Average Layers Used by Input Difficulty**

| Method | Easy | Medium | Hard | Ambiguous |
|---|---|---|---|---|
| Full Depth | 12.0 | 12.0 | 12.0 | 12.0 |
| Entropy | 7.9 | 12.0 | 12.0 | 12.0 |
| Patience | 5.8 | 9.2 | 12.0 | 11.4 |
| Power Metric | 4.2 | 7.6 | 12.0 | 12.0 |

*Note: Power Metric is the only method that scales correctly across all four difficulty tiers. Entropy exits only easy inputs. Patience catches easy and medium. PM correctly gives 5.3 layers to easy and 11.6 layers to hard.*

The difficulty-aware allocation in Table 2 is the core contribution. Entropy thresholding works well for easy inputs but gives full depth to everything else — it only responds to high absolute confidence, not to confident improvement. The patience method improves on this but still gives hard inputs full depth (12.0 layers on average). The power metric is the only method that gives hard inputs meaningfully fewer than full layers (11.6) while easy inputs exit at 5.3 — a monotonically appropriate allocation that matches the actual difficulty distribution.

The ambiguous inputs show an interesting pattern: the power metric exits them at 8.4 layers on average, well before full depth. This is correct behavior: ambiguous inputs do not benefit from additional layers — the model's confidence never rises cleanly. P(t) detects the absence of consistent improvement and allows exit. The small accuracy cost (0.3%) comes from cases where the full-depth prediction would have been marginally different from the exit prediction on these uncertain inputs.

## 4.4 Visualization

*Figure 1. Row 1: Example confidence trajectory per difficulty type with P(t) overlay and exit points for each method. Row 2: Exit layer histograms per method. Row 3: Accuracy-efficiency frontier, layers by difficulty, LIF correspondence panel.*

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# 5. Position in the Efficiency Stack

This paper adds a third layer to the efficiency stack introduced in Paper 6. The three layers are genuinely orthogonal:

- **Layer 1 — FlashAttention (Paper 6):** Optimizes memory bandwidth within each attention operation. Saves 15-78% per operation depending on sequence length.
- **Layer 2 — Power Metric, run-level (Papers 1-2):** Reduces which training and inference runs happen at all. Saves 44-81% of total runs.
- **Layer 3 — Early Exit (this paper):** Reduces the depth of each forward pass. Saves 28.3% of layer computations with 0.985 confidence ratio.

Combined savings at sequence length 4K: FlashAttention saves 65% per operation. Early exit reduces depth by 28.3% at comparable accuracy to full pass. Run-level power metric eliminates 80.7% of inference calls. Total remaining compute: (1-0.65)(1-0.357)(1-0.807) = 4.3%. A system using all three optimizations uses less than 5% of the compute of a baseline system using none.

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## 6. Limitations

5. **Simulated trajectories:** Confidence trajectories are synthetic. Real transformer confidence signals are noisier, can decrease at intermediate layers, and vary substantially across model architectures, tasks, and input types. Validation requires trained exit classifiers on real models (BERT, GPT-2, LLaMA).
6. **Training overhead:** Early exit requires exit classifiers at each layer. These must be trained jointly with the backbone or via distillation. The power metric as exit criterion is a drop-in replacement for the exit decision rule, but the classifiers still require training.
7. **Layer vs run timescales:** The power metric parameters (α, λ, EWMA span) are calibrated to training-run timescales (tens of checkpoints). At the layer level (12 steps), shorter memory (higher α, λ) may be optimal. Ablation study on layer-level parameters is needed.
8. **Accuracy-compute tradeoff:** The 0.985 confidence ratio at 28.3% savings reflects near-full-pass quality. for some applications and not others. A sweep over threshold values would characterize the full Pareto frontier, which is left for future work.

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## 7. Conclusion

We have proposed applying the power metric as the exit criterion for early exit transformers. The signal tracks confidence improvement velocity relative to an adaptive expected trajectory, producing difficulty-aware layer allocation that no static method achieves: 5.3 layers for easy inputs, 11.6 for hard. The mathematical foundation is exact — this is the LIF neuron model applied at the layer level, with the same exponential decay kernel proven optimal in prior theoretical work (Cantrell 2026).

In a simulation calibrated to BERT-base architecture, the power metric achieves 55.9% compute savings with 99.7% accuracy preservation — more savings than confidence thresholding (14.6%) or the patience method (52.6%), with comparable accuracy. When combined with FlashAttention and run-level power metric allocation, the three orthogonal optimizations reduce total inference compute to less than 5% of baseline at standard sequence lengths.

The patience mechanism (BERT Loses Patience) is the closest prior work and the natural comparison. The power metric generalizes it: W(t) alone is approximately the patience mechanism, while E(t)×W(t) adds the efficiency signal that calibrates the exit to the input's actual improvement trajectory. This is the same addendum relationship that Paper 7 identified with Hyperband: not a replacement, but a principled extension that addresses a specific limitation of the existing method.

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### Contact & Collaboration


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The natural experiment: implement power metric exit criterion on a BERT-base model with standard exit classifiers and test on SST-2, MNLI, and SQuAD. Compare directly against BERT Loses Patience (Zhou et al. 2020) and entropy thresholding. Framework, simulation code, and patent application available on request.